

Reasoning, Acting, and Fallback: A Reproduction Study on HotpotQA and FEVER

ReAct ICLR 2023 Reproduction Project

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Abstract

We evaluate seven prompting and agent strategies on matched 500-example subsets of HotpotQA and FEVER using Qwen2.5-7B-Instruct. The study covers Standard prompting, chain-of-thought (CoT), self-consistent chain-of-thought (CoT-SC), Act, ReAct, and two directional hybrids. On HotpotQA, ReAct followed by CoT-SC obtains the strongest answer performance, reaching 31.4 exact match (EM) and 43.54 F1. On FEVER, the same hybrid reaches 61.6 accuracy and produces no invalid final labels. Trajectory analysis shows that completed ReAct decisions are comparatively strong, but standalone ReAct is constrained by retrieval formulation and failure to finish within the interaction budget. The contrast is particularly marked on FEVER: completed ReAct trajectories achieve 67.8 accuracy, yet 258 of 500 examples return no valid label. These results support the use of fallback policies while also showing that their effectiveness depends on direction, task structure, and the cost of the route attempted first.

1 Introduction

ReAct interleaves natural-language reasoning with actions in an external environment, providing a mechanism by which a language model can retrieve information and revise its reasoning as evidence arrives [1]. This design is attractive for knowledge-intensive tasks because it joins parametric knowledge with explicit interaction. It also creates a control problem: a model must formulate a useful query, parse the observation, decide whether more evidence is required, and finish before its action budget is exhausted.

This report studies that trade-off on two tasks used in the original ReAct work. HotpotQA is open-ended multi-hop question answering, whereas FEVER is three-way claim verification. Although both depend on factual knowledge, their output spaces and failure modes differ. A HotpotQA system must compose an answer string after following multiple relations. A FEVER system must instead map evidence to one of `extscSupports`, `extscRefutes`, or `extscNot Enough Info`.

All quantitative results are extracted from the benchmark artifacts currently stored in `outputs/hotpotqa/` and `outputs/fever/`. The report does not estimate missing values or merge smoke tests with full runs. The accompanying deterministic utility selects and validates runs, produces machine-readable summaries, and generates the figures used below.

2 Background

2.1 ReAct

ReAct represents each interaction as a Thought, an Action, and an environment Observation. In this implementation, the action space consists of `Search[query]`, `Lookup[term]`, and `Finish[answer]`. Search and Lookup observations come only from the Wikipedia environment; model-generated Observation text is stopped and deterministically truncated before action parsing.

2.2 Chain-of-Thought

Chain-of-thought prompting elicits an intermediate reasoning trace before a final answer [2]. It provides no external evidence in these experiments, so success depends on the model’s parametric knowledge and reasoning. Standard prompting provides the corresponding direct-answer baseline.

2.3 Self-Consistency

Self-consistency samples multiple reasoning paths and aggregates their normalized answers [3]. The runs use 21 samples at temperature 0.7. FEVER voting occurs over three canonical labels; HotpotQA voting occurs over normalized open-ended strings, creating a much larger answer space.

2.4 Hybrid Reasoning Strategies

The two hybrids are directional. $\text{ReAct} \rightarrow \text{CoT-SC}$ keeps a completed ReAct answer and calls CoT-SC only after ReAct fails. $\text{CoT-SC} \rightarrow \text{ReAct}$ accepts the vote when the winning count exceeds $n/2$ and otherwise starts ReAct. With $n = 21$, the threshold is 10.5, so at least 11 votes are required. The names and arrows in this report follow the implemented execution order.

3 Experimental Setup

3.1 Model

Every selected run uses Qwen/Qwen2.5-7B-Instruct. FEVER pins model revision `a09a35458c70` (the full SHA is preserved in the machine-readable summary); the older HotpotQA configs do not record a model revision. Deterministic legs use temperature 0.0, top-p 1.0, and at most 256 new tokens. Batch size varies by method.

3.2 HotpotQA

The HotpotQA runs use `hotpotqa/hotpot_qa:distractor:validation`, seed 42, 500 examples, and a seven-step agent budget. Answer EM is the primary metric, with token-level F1 as a secondary metric. The agents do not output supporting-fact annotations, so the stored supporting-fact and joint metrics are zero and are excluded from the main comparison.

3.3 FEVER

The FEVER runs use the `paper_dev` subset and `dev` split of `ysmyth/ReAct`, pinned to revision `6bdb3a1fd38b` (full SHA in the summary). The sample contains 500 claims selected with seed 42. The agent budget is five steps, and accuracy over the three canonical labels is the primary metric.

3.4 Wikipedia Environment

Search opens an exact article when possible and otherwise returns suggestions; Lookup scans the active article for successive occurrences. Suggestions are observations, not actions: the environment does not automatically replace a failed query with the first suggested title. This distinction is central to the FEVER failures analyzed below.

3.5 Evaluated Methods

We evaluate Standard, CoT, CoT-SC, Act, ReAct, $\text{ReAct} \rightarrow \text{CoT-SC}$, and $\text{CoT-SC} \rightarrow \text{ReAct}$. Standard and CoT use one completion per example. CoT-SC uses 21. Act omits explicit thoughts but interacts with Wikipedia. ReAct combines explicit reasoning and actions. The hybrids retain the same component settings and routing policies.

3.6 Evaluation Metrics

HotpotQA uses normalized answer EM and F1. FEVER uses label accuracy, invalid/unparsed count, class-conditional accuracy, and a confusion matrix. For interactive methods, the artifacts also record logical steps,

Wikipedia tool calls, termination reasons, and trajectories. Runtime is wall-clock time from each run and is not hardware-normalized.

3.7 Reproducibility Configuration

A run is selected only if its task and method metadata match its directory, its configured sample count matches its metrics and prediction records, all example identifiers are unique, and its log reaches the final metrics write without an error. The selector prefers the largest run and then the latest timestamp. All selected runs contain 500 examples, and the example order is identical across methods within each task.

The FEVER suite is internally consistent at code version 0.11.0. HotpotQA is historical: four methods use 0.10.0, the hybrids use 0.9.0, and standalone ReAct uses 0.8.1. This limits causal interpretation of small HotpotQA differences.

4 HotpotQA Results

4.1 Main Results

Table 1: HotpotQA results on the selected 500-example runs.

Method	Correct	EM	F1	Steps	Tools	Runtime (min)
Standard	92	18.4	27.74	1.000	0.000	0.42
CoT	108	21.6	30.63	1.000	0.000	2.44
CoT-SC	118	23.6	33.37	21.000	0.000	54.16
Act	135	27.0	36.43	3.534	2.456	22.99
ReAct	129	25.8	34.49	5.144	4.352	195.66
ReAct $\rightarrow$ CoT-SC	157	31.4	43.54	15.250	4.332	131.99
CoT-SC $\rightarrow$ ReAct	146	29.2	38.38	24.138	2.678	235.86

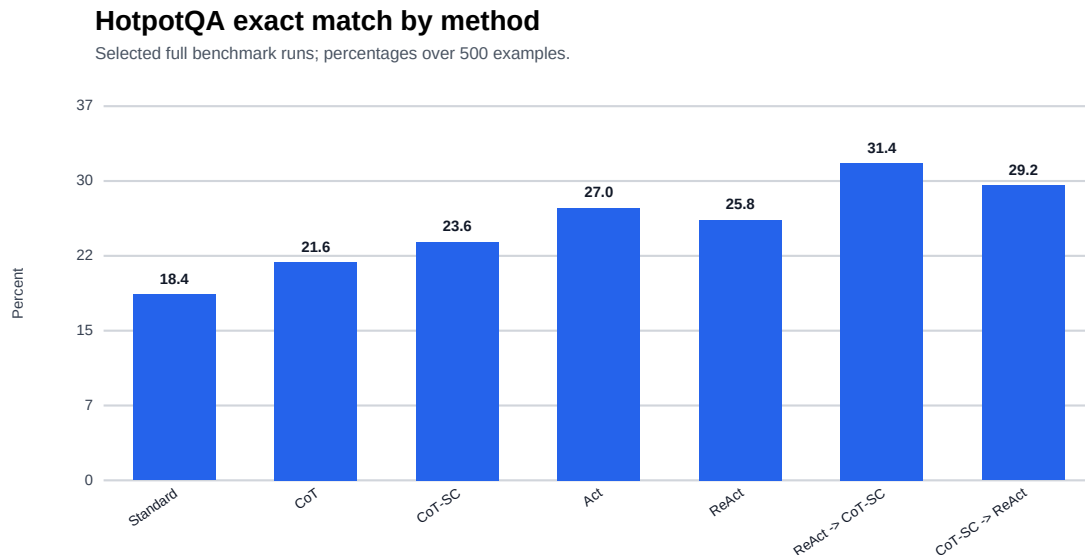


Figure 1: HotpotQA exact match. The figure is generated from `results_summary.json`.

ReAct $\rightarrow$ CoT-SC has the highest EM and F1. Act is the strongest single interactive strategy at 27.0 EM, slightly ahead of ReAct at 25.8. The closed-book sequence Standard, CoT, and CoT-SC improves monotonically, but each increment increases inference cost.

4.2 Agent and Retrieval Behavior

Standalone ReAct completes 279 trajectories, exhausts the step budget on 197, and stops on an action loop on 24. Its 129 exact matches all occur among completed trajectories, yielding 46.2% EM within that subset. Across the run, ReAct makes 1,811 searches and 480 lookups. It records 346 Lookup misses and repeats an exact Search action in 116 examples.

ReAct $\rightarrow$ CoT-SC preserves 260 completed ReAct results, of which 122 are exact matches. Its 240 fallback examples add 35 exact matches. CoT-SC $\rightarrow$ ReAct accepts 216 initial consensus results, with 102 exact matches; ReAct completes 128 of the 284 fallback cases and answers 44 correctly, while 156 hybrid trajectories fail to produce an answer.

4.3 Failure Analysis

The trajectories distinguish failures to identify an entity from failures to retrieve the second-hop relation. In one successful case, ReAct opens Greyia and Calibanus, compares their reported species counts, and answers Greyia in three steps. In the Tokarev failure, the agent correctly discovers Moscow State University but cannot retrieve its founding year. Two unsuccessful lookups and three reformulated searches consume the remaining budget. This is not an answer-parser failure; it is a retrieval-control failure ending in `max_steps_exceeded`.

The run contains only two malformed standalone ReAct action steps, whereas 221 examples end through a step limit or loop. Formatting therefore has little explanatory power for the aggregate HotpotQA failure rate. The larger issue is conversion of partial evidence into an efficient second-hop query and a timely Finish action.

5 FEVER Results

5.1 Main Results

Table 2: FEVER results on the selected 500-example runs.

Method	Correct	Accuracy	Invalid	Steps	Tools	Runtime (min)
Standard	258	51.6	2	1.000	0.000	1.31
CoT	280	56.0	5	1.000	0.000	2.93
CoT-SC	286	57.2	0	21.000	0.000	62.93
Act	271	54.2	39	2.806	1.802	27.98
ReAct	164	32.8	258	4.058	3.432	79.42
ReAct $\rightarrow$ CoT-SC	308	61.6	0	15.064	3.446	170.95
CoT-SC $\rightarrow$ ReAct	275	55.0	15	21.216	0.190	118.03

ReAct $\rightarrow$ CoT-SC reaches 61.6% accuracy and has no invalid final predictions. CoT-SC is the strongest non-hybrid at 57.2%. Standalone ReAct returns 258 invalid/unparsed predictions, causing its aggregate accuracy to fall to 32.8% despite strong completed cases.

5.2 Class-Level Analysis

Table 3: FEVER per-class accuracy (%).

Method	Supports	Refutes	Not Enough Info
Standard	55.7	74.6	29.9
CoT	75.9	68.3	27.7
CoT-SC	78.7	68.3	28.3
Act	52.9	73.2	40.8
ReAct	37.9	35.9	25.5
ReAct $\rightarrow$ CoT-SC	78.7	70.4	38.6
CoT-SC $\rightarrow$ ReAct	75.3	66.9	26.6

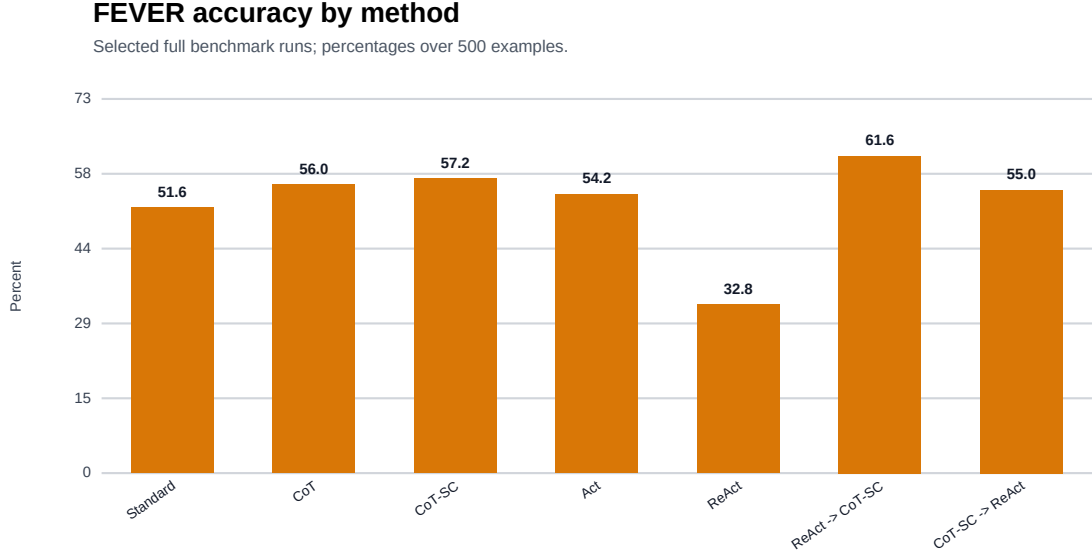


Figure 2: FEVER label accuracy. The large standalone ReAct deficit includes non-completions.

Not Enough Info is the hardest class for every method. Self-consistency improves Supports but leaves NEI below 29%. Act obtains the highest NEI accuracy, although its invalid predictions and weaker Supports performance lower its aggregate score. ReAct $\rightarrow$ CoT-SC combines strong Supports and Refutes performance with a substantial NEI improvement over standalone CoT-SC.

5.3 Agent and Retrieval Behavior

Standalone ReAct completes 242 examples, reaches the five-step limit on 243, and ends with a parsing error on 15. All 164 correct predictions are in the completed group, which has 67.8% accuracy. Empty predictions from budget exhaustion must therefore be separated from malformed generations.

ReAct makes 1,705 searches, of which 1,469 receive a **Could not find** observation with suggestions. Only 11 Lookup actions appear. For the Peking University claim, five increasingly specific queries all return suggestions containing the exact article title, but the model never issues **Search[Peking University]**. The environment preserves paper-style action semantics rather than silently selecting that suggestion.

The current implementation attempts one action-only recovery when a combined Thought/Action completion cannot be parsed. The retry occurs in the same logical step, runs only for failed batch states, stops at a newline, and is subjected to deterministic truncation. The serializer stores the recovered completion rather than both generations. In standalone FEVER, 29 action-only serialized outputs are consistent with successful recovery, while 71 steps across 68 examples remain invalid after recovery. This modern batched implementation preserves the original protocol’s same-step semantics but batches the recovery calls for efficiency and state isolation.

5.4 Failure Analysis

The artifacts expose three quantitatively important classes of standalone ReAct failure. First, 243 examples exhaust the environment budget without Finish. Second, 15 finish with **parsing_error**; malformed steps can also occur earlier in trajectories that continue. Third, 78 of the 242 completed examples return a valid but wrong label, indicating evidence selection or interpretation errors. These categories should not be collapsed into a single parser-failure explanation.

Act has 39 invalid predictions, with 409 ordinary completions, 52 completions at the step limit, and 39 failed finalizations. Standard and CoT have 2 and 5 invalid labels, respectively; CoT-SC has none. The label parser is therefore not the dominant source of FEVER error outside a small number of direct-generation cases.

6 Cross-Task Comparison

HotpotQA and FEVER both benefit from factual knowledge, but they stress different aspects of the agent loop. HotpotQA requires open-ended answer normalization and often an explicit second hop. Its successful trajectories commonly retrieve two entities and compose a relation across them. FEVER has a small output vocabulary, but the agent must determine whether evidence entails, contradicts, or fails to resolve a claim.

This difference affects self-consistency. On FEVER, 21 generations vote over three labels and always yield a valid aggregate label in the selected run. On HotpotQA, surface variation fragments votes across many strings, and a normalized plurality can still be weak. Retrieval also behaves differently: HotpotQA Lookup is used frequently, while FEVER ReAct overwhelmingly issues Search queries and rarely follows with Lookup.

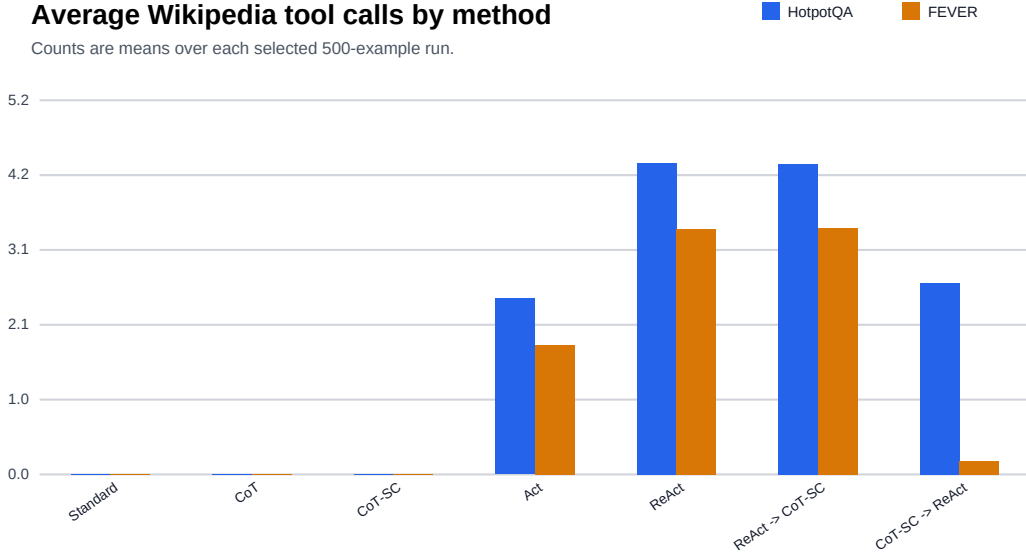


Figure 3: Average Wikipedia tool calls. CoT-SC steps are not tool calls and are excluded from this accounting.

The hybrid direction is consequential on both tasks. ReAct $\rightarrow$ CoT-SC performs best overall because a completed ReAct path can be retained while the closed-book vote supplies an answer for failures. CoT-SC $\rightarrow$ ReAct uses few FEVER tools because only 25 claims fall below its confidence threshold; however, those cases are unusually difficult, and ReAct completes only 10 of them.

7 Discussion

7.1 ReAct versus Non-Agentic Reasoning

Interactive methods reduce dependence on parametric knowledge but add exposure to query formulation, environment state, and termination. On HotpotQA, Act and ReAct outperform Standard, CoT, and CoT-SC in EM. On FEVER, completed ReAct cases are accurate, but its low completion rate overwhelms that advantage. ReAct should therefore be evaluated jointly on decision quality and completion behavior.

7.2 Effect of Self-Consistency

CoT-SC improves over CoT by 2.0 EM points on HotpotQA and 1.2 accuracy points on FEVER. It also removes invalid FEVER labels. The gains are measurable but expensive: 21 generations per example increase runtime from 2.44 to 54.16 minutes on HotpotQA and from 2.93 to 62.93 minutes on FEVER.

7.3 Hybrid Reasoning Strategies

ReAct $\rightarrow$ CoT-SC improves robustness by filling in missing ReAct answers. On FEVER, its completed-ReAct path has 68.5% accuracy across 238 examples, while the fallback path has 55.3% across 262. CoT-SC $\rightarrow$ ReAct accepts 475 FEVER votes at 57.1% accuracy; its 25 fallback cases yield only 4 correct labels and 15 failures. Routing is therefore not neutral: the first-stage confidence rule determines the difficulty distribution seen by the fallback.

7.4 Accuracy versus Tool and Inference Cost

Standard and CoT are inexpensive but less accurate. CoT-SC spends sampling compute without tools. ReAct spends environment calls and may still fail to answer. ReAct $\rightarrow$ CoT-SC has the strongest scores but the highest FEVER runtime because many examples pay for both a full ReAct attempt and 21 CoT samples. Wall-clock values remain descriptive because batch sizes differ.

7.5 Effect of Using a Modern Instruction-Tuned Model

Table 4 separates the original PaLM-540B values in the repository’s copy of the ReAct paper from the present Qwen reproduction. The comparison is not model-controlled. It is included to locate the reproduction, not to attribute differences to a single cause.

Table 4: Original-paper results and this reproduction (percentage points).

Method	HotpotQA EM		FEVER accuracy	
	Paper	Reproduction	Paper	Reproduction
Standard	28.7	18.4	57.1	51.6
CoT	29.4	21.6	56.3	56.0
CoT-SC	33.4	23.6	60.4	57.2
Act	25.7	27.0	58.9	54.2
ReAct	27.4	25.8	60.9	32.8
ReAct $\rightarrow$ CoT-SC	35.1	31.4	62.0	61.6
CoT-SC $\rightarrow$ ReAct	34.2	29.2	64.6	55.0

The absolute gaps vary across methods and tasks. In particular, the FEVER standalone ReAct score is governed by completion failures specific to this model–environment interaction. At the same time, the ReAct $\rightarrow$ CoT-SC FEVER result is close to the original paper value. These observations suggest that the protocol’s fallback structure transfers more reliably than every component’s standalone performance.

8 Limitations

The experiments use 500-example seeded samples rather than complete evaluation splits, and no confidence intervals or significance tests are reported. HotpotQA methods span multiple historical code versions, and its model revision and standalone ReAct batch size are missing. Wikipedia is a live external dependency without a pinned content snapshot. Runtimes are affected by batching and hardware. HotpotQA supporting-fact metrics are unavailable because supporting facts are not predicted. Finally, the original-paper comparison changes the model, inference stack, and likely environment state, so it is descriptive rather than causal.

9 Conclusion

Across both tasks, ReAct $\rightarrow$ CoT-SC provides the strongest aggregate result: 31.4 EM on HotpotQA and 61.6 accuracy on FEVER. The advantage arises from complementarity. ReAct produces strong answers when it completes, while CoT-SC supplies a valid answer when interaction fails. The trajectory evidence also identifies the remaining bottleneck: retrieval access alone is insufficient when the model formulates over-specific queries, ignores useful suggestions, or does not emit a final answer before the environment budget expires.

A Full Configuration

Table 5: Selected-run configuration. A dash denotes a field not recorded or not applicable.

Task / method	Code	Batch	Max steps	CoT-SC n	CoT-SC temp.	Best effort
HotpotQA Standard	0.10.0	32	7	–	–	–
HotpotQA CoT	0.10.0	16	7	–	–	–
HotpotQA CoT-SC	0.10.0	16	7	21	0.7	–
HotpotQA Act	0.10.0	16	7	–	–	–
HotpotQA ReAct	0.8.1	–	7	–	–	false
HotpotQA ReAct → CoT-SC	0.9.0	8	7	21	0.7	false
HotpotQA CoT-SC → ReAct	0.9.0	3	7	21	0.7	false
FEVER Standard	0.11.0	32	5	–	–	–
FEVER CoT	0.11.0	16	5	–	–	–
FEVER CoT-SC	0.11.0	16	5	21	0.7	–
FEVER Act	0.11.0	16	5	–	–	–
FEVER ReAct	0.11.0	8	5	–	–	false
FEVER ReAct → CoT-SC	0.11.0	8	5	21	0.7	false
FEVER CoT-SC → ReAct	0.11.0	8	5	21	0.7	false

B Additional Metrics

Table 6: Termination and route counts used in the analysis.

Run	Counts
HotpotQA ReAct	completed 279; max steps 197; action loop 24
HotpotQA ReAct → CoT-SC	ReAct completed 260; CoT-SC fallback 240
HotpotQA CoT-SC → ReAct	CoT-SC consensus 216; ReAct completed 128; failed 156
FEVER ReAct	completed 242; max steps 243; parsing error 15
FEVER ReAct → CoT-SC	ReAct completed 238; CoT-SC fallback 262
FEVER CoT-SC → ReAct	CoT-SC consensus 475; ReAct completed 10; failed 15

C Representative Trajectories

Successful HotpotQA multi-hop ReAct. For the Greyia–Calibanus comparison, the abbreviated sequence is `Search[Greyia]` → “three species,” `Search[Calibanus]` → “two species,” then `Finish[Greyia]`.

HotpotQA failure. For the Tokarev question, the agent identifies Moscow State University but receives no match for two founding-year lookups. Three reformulated searches return the same lead, and step seven ends without `Finish`.

Successful FEVER ReAct. For “Yandex is an application,” `Search[Yandex]` opens an article describing a technology company and its services. The next action is `Finish[REFUTES]`.

FEVER max-step failure. For the Peking University founding-century claim, five over-specific searches fail while suggesting the exact title. ReAct never follows the suggestion and exhausts the budget.

Hybrid fallback. ReAct $\rightarrow$ CoT-SC applies the same five unsuccessful Peking University searches, then samples 21 CoT decisions. The vote consistently identifies 1898 as a nineteenth-century founding date and returns REFUTES.

References

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